Review of Basics

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Outline

- Sobolev Spaces
- Model Problems
- Finite Element Spaces
- *A Priori* Analysis
- *A Posteriori* Analysis
- Variational Crimes
- Miscellaneous Topics

General References

1. S.C. Brenner and L.R. Scott
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   Finite Element Methods, in Encyclopedia of Computational Mechanics, pages 73–118, Wiley, 2004.

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Sobolev Spaces

$\Omega \subset \mathbb{R}^d$ is a bounded open subset with a reasonable boundary.

$H^m(\Omega) =$ space of functions whose weak derivatives up to order m are square integrable.

$H^m(\Omega)$ is a Hilbert space with the inner product

$$(v, w)_{H^m(\Omega)} = \sum_{|\alpha| \le m} \int_\Omega \left(\frac{\partial^{|\alpha|} v}{\partial x^\alpha}\right)\left(\frac{\partial^{|\alpha|} w}{\partial x^\alpha}\right) dx$$

$$\alpha = (\alpha_1, \ldots, \alpha_d), \quad |\alpha| = \alpha_1 + \cdots + \alpha_d, \quad \partial x^\alpha = (\partial x_1)^{\alpha_1} \cdots (\partial x_d)^{\alpha_d}$$

The Sobolev norm $\|\cdot\|_{H^m(\Omega)}$ is given by

$$\|v\|^2_{H^m(\Omega)} = \sum_{|\alpha| \le m} \left\|(\partial^{|\alpha|} v / \partial x^\alpha)\right\|^2_{L_2(\Omega)}$$

Sobolev Spaces

The Sobolev semi-norm $|\cdot|_{H^m(\Omega)}$ is defined by

$$|v|^2_{H^m(\Omega)} = \sum_{|\alpha|=m} \left\| (\partial^{|\alpha|} v / \partial x^\alpha) \right\|^2_{L_2(\Omega)}$$

$H^m_0(\Omega)$ is the closure of $C^\infty_0(\Omega)$ in $H^m(\Omega)$, where $C^\infty_0(\Omega)$ is the space of C^∞ functions on Ω with compact support.

Equivalently, $v \in H^m_0(\Omega)$ if and only if $v \in H^m(\Omega)$ and the trace of $(\partial^{|\alpha|} v / \partial x^\alpha)$ on $\partial\Omega$ vanishes for $|\alpha| \leq m-1$.

Poincaré-Friedrichs Inequalities

$$\|v\|_{L_2(\Omega)} \leq C_\Omega \left(|v|_{H^1(\Omega)} + \left| \int_\Omega v \, dx \right| \right) \qquad \forall\, v \in H^1(\Omega)$$

$$\|v\|_{L_2(\Omega)} \leq C_\Omega \left(|v|_{H^1(\Omega)} + \|v\|_{L_2(\Gamma)} \right) \qquad \forall\, v \in H^1(\Omega)$$

$\Gamma \subset \partial\Omega$ and $|\Gamma| > 0$

Sobolev Spaces

Sobolev's Embedding Theorem For any $v \in H^m(\Omega)$ where $2m > d$, there exists $v_\dagger \in C(\bar{\Omega})$ such that $v = v_\dagger$ almost everywhere and

$$\sup_{x \in \bar{\Omega}} |v_\dagger(x)| \leq C \|v\|_{H^m(\Omega)}$$

($v_\dagger$ is usually identified with v and hence $H^m(\Omega) \hookrightarrow C(\bar{\Omega})$)

It follows that $H^m(\Omega)$ can be embedded in $C^k(\bar{\Omega})$, i.e.,

$$\|v\|_{C^k(\bar{\Omega})} \leq C \|v\|_{H^m(\Omega)} \qquad \forall\, v \in H^m(\Omega)$$

provided that $2(m-k) > d$. Hence the pointwise evaluation of derivatives of v up to order k is well-defined.

Note that $H^1(\Omega) \not\hookrightarrow C(\bar{\Omega})$ for $\Omega \subset \mathbb{R}^d$ ($d \geq 2$) and the pointwise evaluation of functions in $H^1(\Omega)$ is not well-defined.

Model Problems

$\Omega \subset \mathbb{R}^d \quad (d = 2, 3)$

Diffusion Problem

$$\begin{aligned} -\Delta u &= f \qquad \text{in} \quad \Omega \\ u &= 0 \qquad \text{on} \quad \partial\Omega \end{aligned}$$

Weak Formulation

Find $u \in V = H_0^1(\Omega)$ such that

$$a(u, v) = F(v) \qquad \forall\, v \in V$$

$$a(u, v) = \int_\Omega \nabla u \cdot \nabla v \, dx, \qquad F(v) = \int_\Omega f v \, dx$$

Model Problems

Elasticity Problem

$$-\nabla \cdot \boldsymbol{\sigma}(\boldsymbol{u}) = \boldsymbol{f} \qquad \text{in} \quad \Omega, \qquad \boldsymbol{u} = \boldsymbol{0} \qquad \text{on} \quad \partial\Omega$$

$$\boldsymbol{\sigma}(\boldsymbol{u}) = 2\mu\boldsymbol{\epsilon}(\boldsymbol{u}) + \lambda(\nabla \cdot \boldsymbol{u})\boldsymbol{\delta}, \qquad \boldsymbol{\epsilon}(\boldsymbol{u}) = \frac{1}{2}\left[\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^t\right]$$

$\boldsymbol{\delta} =$ identity matrix

Weak Formulation

Find $\boldsymbol{u} \in V = [H_0^1(\Omega)]^d$ such that

$$a(\boldsymbol{u}, \boldsymbol{v}) = F(\boldsymbol{v}) \qquad \forall\, \boldsymbol{v} \in V$$

$$a(\boldsymbol{u}, \boldsymbol{v}) = \int_\Omega \left[2\mu\boldsymbol{\epsilon}(\boldsymbol{u}) : \boldsymbol{\epsilon}(\boldsymbol{v}) + \lambda(\nabla \cdot \boldsymbol{u})(\nabla \cdot \boldsymbol{v})\right] dx$$

$$F(\boldsymbol{v}) = \int_\Omega \boldsymbol{f} \cdot \boldsymbol{v}\, dx$$

Model Problems

Biharmonic Problem

$$\Delta^2 u = f \qquad \text{in} \quad \Omega$$

$$u = 0 \qquad \text{on} \quad \partial\Omega, \qquad \frac{\partial u}{\partial n} = 0 \qquad \text{on} \quad \partial\Omega$$

Weak Formulation

Find $u \in V = H_0^2(\Omega)$ such that

$$a(u,v) = F(v) \qquad \forall\, v \in V$$

$$a(u,v) = \int_\Omega (\Delta u)(\Delta v) dx \quad \text{or} \quad a(u,v) = \int_\Omega \nabla^2 u : \nabla^2 v \, dx$$

$$F(v) = \int_\Omega fv \, dx$$

Model Problems

Stokes Problem

$$-\Delta \boldsymbol{u} + \nabla p = \boldsymbol{f} \quad \text{in} \quad \Omega, \qquad \boldsymbol{u} = \boldsymbol{0} \quad \text{on} \quad \partial\Omega$$

$$\nabla \cdot \boldsymbol{u} = 0 \quad \text{in} \quad \Omega \qquad \left(p \in \mathring{L}_2(\Omega) = \left\{ q \in L_2(\Omega) : \int_\Omega q \, dx = 0 \right\} \right)$$

Weak Formulation

Find $\boldsymbol{u} \in V = [H_0^1(\Omega)]^d$ and $p \in Q = \mathring{L}_2(\Omega)$ such that

$$\begin{aligned} a(\boldsymbol{u}, \boldsymbol{v}) + b(\boldsymbol{v}, p) &= F(\boldsymbol{v}) \qquad \forall \boldsymbol{v} \in V \\ b(\boldsymbol{u}, q) &= G(q) \qquad \forall q \in Q \end{aligned}$$

$$a(\boldsymbol{u}, \boldsymbol{v}) = \int_\Omega \nabla \boldsymbol{u} : \nabla \boldsymbol{v} \, dx, \qquad b(\boldsymbol{v}, p) = -\int_\Omega (\nabla \cdot \boldsymbol{v}) p \, dx$$

$$F(v) = \int_\Omega \boldsymbol{f} \cdot \boldsymbol{v}, \qquad G(q) = 0$$

Model Problems

Symmetric Positive Definite Problem (diffusion, elasticity, biharmonic)

Find $u \in V$ such that

$$a(u, v) = F(v) \qquad \forall\, v \in V$$

$V =$ Hilbert space
$a(\cdot, \cdot) =$ SPD bilinear form on $V \times V$, $\quad F \in V'$

Ritz-Galerkin Method

Find $\tilde{u} \in \tilde{V}$ such that

$$a(\tilde{u}, v) = F(v) \qquad \forall\, v \in \tilde{V}$$

$\tilde{V} =$ finite dimensional subspace of V

Model Problems

Saddle Point Problem (Stokes)

Find $(u, p) \in V \times Q$ such that

$$\begin{aligned} a(u, v) + b(v, p) &= F(v) \qquad \forall\, v \in V \\ b(u, q) &= G(q) \qquad \forall\, q \in Q \end{aligned}$$

$V, Q =$ Hilbert spaces
$a(\cdot, \cdot) =$ bilinear form on $V \times V$, $\quad b(\cdot, \cdot) =$ bilinear form on $V \times Q$
$F \in V', \quad G \in Q'$

Ritz-Galerkin Method

Find $(\tilde{u}, \tilde{p}) \in \tilde{V} \times \tilde{Q}$ such that

$$\begin{aligned} a(\tilde{u}, v) + b(\boldsymbol{v}, \tilde{p}) &= F(v) \qquad \forall\, v \in \tilde{V} \\ b(\tilde{u}, q) &= G(q) \qquad \forall\, q \in \tilde{Q} \end{aligned}$$

$\tilde{V} =$ finite dimensional subspace of V
$\tilde{Q} =$ finite dimensional subspace of Q

Finite Element Spaces

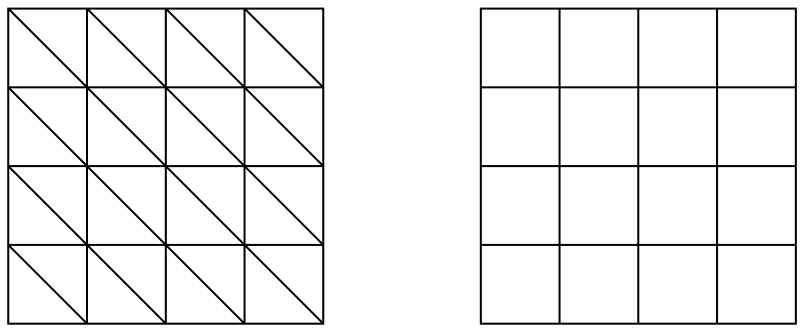
Finite element methods are Ritz-Galerkin methods where the finite dimensional subspaces are constructed by gluing together polynomials defined on subdomains.

For simplicity we will focus on the construction of finite element spaces in two dimensions.

Finite Element Spaces

Triangulations

Let $\Omega \subset \mathbb{R}^2$ be a bounded polygonal domain. A triangulation of Ω is a partition of Ω into triangles or quadrilaterals.



The intersection of two distinct closed triangles (quadrilaterals) is either (i) empty, (ii) a common vertex, or (iii) a common edge.

The mesh size h of a triangulation is the maximum of the diameters of the triangles (quadrilaterals).

Finite Element Spaces

A family of triangulations $\mathcal{T}_h$ is regular (or non-degenerate) if there exists a constant $\rho > 0$ such that

$$\operatorname{diam} B_T \geq \rho h_T \qquad \forall\, T \in \mathcal{T}_h$$

where B_T is the largest ball inside T and h_T is the diameter of T.

$\mathcal{T}_h$ is quasi-uniform if there exists a constant $\rho > 0$ such that

$$\operatorname{diam} B_T \geq \rho h \qquad \forall\, T \in \mathcal{T}_h$$

The finite element space V_h is constructed by gluing together polynomials defined on the triangles (quadrilaterals) in $\mathcal{T}_h$.

Finite Element Spaces

Finite Element $(\mathcal{K}, \mathcal{P}, \mathcal{N})$

$\mathcal{K} =$ a closed triangle (quadrilateral)

$\mathcal{P} =$ a finite dimensional vector space of shape functions

$\mathcal{N} =$ a basis of $\mathcal{P}'$

The linear functionals in $\mathcal{N}$ are known as nodal variables (or degrees of freedom).

$\mathcal{N} = \{N_1, \ldots, N_m\}$ ($m = \dim \mathcal{P}$) is a basis of $\mathcal{P}'$ if and only if any function $v \in \mathcal{P}$ is uniquely determined by the values $N_1(v), \ldots, N_m(v)$.

Equivalently, $\mathcal{N}$ is a basis of $\mathcal{P}'$ if and only if

$$v \in \mathcal{P} \quad \text{and} \quad N_1(v) = \cdots = N_m(v) = 0 \Longrightarrow v = 0$$

Finite Element Spaces

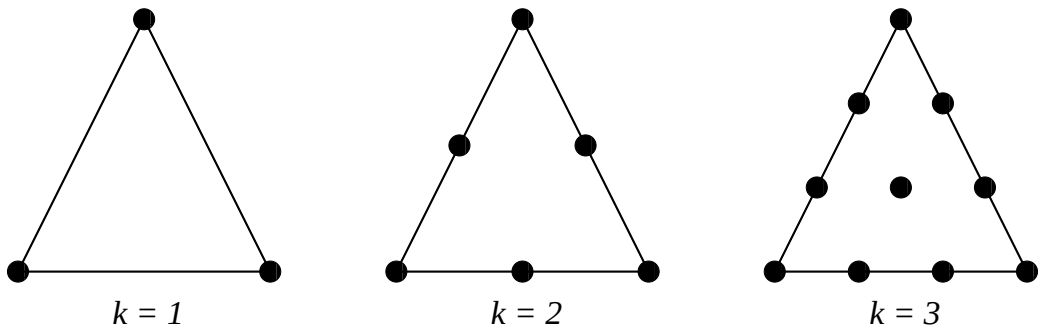
Finite Element $(\mathcal{K}, \mathcal{P}, \mathcal{N})$

$\mathcal{K} =$ a closed triangle (quadrilateral)

$\mathcal{P} =$ a finite dimensional vector space of shape functions

$\mathcal{N} =$ a basis of $\mathcal{P}'$

$\mathbb{P}_k$ Lagrange Finite Elements



$\mathcal{K} =$ triangle

$\mathcal{P} = \mathbb{P}_k =$ space of polynomials of total degree $\leq k$

$\mathcal{N} =$ evaluations at the $(k+1)(k+2)/2$ nodes

Finite Element Spaces

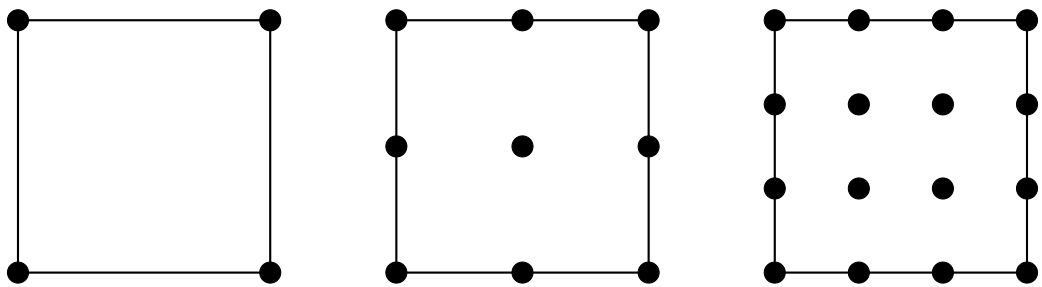
Finite Element $(\mathcal{K}, \mathcal{P}, \mathcal{N})$

$\mathcal{K} =$ a closed triangle (quadrilateral)

$\mathcal{P} =$ a finite dimensional vector space of shape functions

$\mathcal{N} =$ a basis of $\mathcal{P}'$

$\mathbb{Q}_k$ Lagrange Finite Elements



$\mathcal{K} =$ rectangle

$\mathcal{P} = \mathbb{Q}_k =$ space of polynomials of individual degree $\leq k$

$\mathcal{N} =$ evaluations at the $(k+1)^2$ nodes

Finite Element Spaces

$\mathbb{P}_k$ Lagrange Finite Element Space

$\mathcal{T}_h =$ triangulation of Ω by triangles

$V_h =$ finite element space obtained by gluing together piecewise polynomials of degree $\leq k$

$v \in V_h$ if and only if

(i) $v_T = v\big|_T$ belongs to $\mathbb{P}_k$ for all $T \in \mathcal{T}_h$

(ii) $v_{T_1} = v_{T_2}$ at the common nodes of T_1 and T_2

$\mathbb{P}_k$ Lagrange finite elements are C^0 elements because functions in V_h are globally continuous.

$V_h \subset H^1(\Omega)$ since the finite element functions in V_h are piecewise smooth and globally continuous.

$\mathbb{Q}_k$ Lagrange finite element space can be similarly defined and have similar properties.

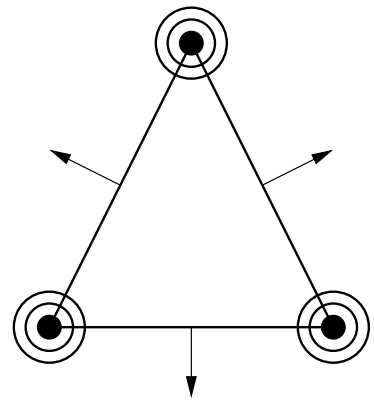
Finite Element Spaces

C^1 Finite Elements

Argyris Element

$$\mathcal{K} = \text{triangle}$$

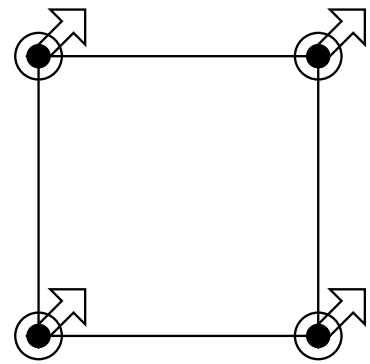
$$\mathcal{P} = \mathbb{P}_5$$



Bogner-Fox-Schmit Element

$$\mathcal{K} = \text{rectangle}$$

$$\mathcal{P} = \mathbb{Q}_3$$



Functions in the finite element space V_h obtained by gluing the Argyris (B-F-S) elements together are globally C^1. Therefore V_h is a subspace of $H^2(\Omega)$.

SPD Problems

Continuous Problem Find $u \in V$ such that

$$a(u, v) = F(v) \quad \forall\, v \in V$$

Stability Conditions There exist $C > c > 0$ such that

- $|a(w, v)| \leq C\|w\|_V \|v\|_V$ for all $v, w \in V$ (bounded)
- $a(v, v) \geq c\|v\|_V^2$ for all $v \in V$ (coercive)

Under these conditions the continuous problem has a unique solution by the Riesz Representation Theorem.

Finite Element Method Find $u_h \in V_h \subset V$ such that

$$a(u_h, v) = F(v) \quad \forall\, v \in V_h$$

Galerkin Orthogonality

$$a(u - u_h, v) = 0 \qquad \forall\, v \in V_h$$

Pythagoras' Theorem

For any $v \in V_h$

$$\begin{aligned} a(u - v, u - v) &= a((u - u_h) + (u_h - v), (u - u_h) + (u_h - v)) \\ &= a(u - u_h, u - u_h) + a(u_h - v, u_h - v) \end{aligned}$$

$$\|u - v\|_a^2 = \|u - u_h\|_a^2 + \|u_h - v\|_a^2$$

The energy norm $\|\cdot\|_a = \sqrt{a(\cdot,\cdot)}$ is equivalent to the (Sobolev) norm of V.

Abstract Error Estimate

$$\|u - u_h\|_a = \min_{v \in V_h} \|u - v\|_a$$

Approximation Property of V_h

Let V_h be either the $\mathbb{P}_k$ or the $\mathbb{Q}_k$ Lagrange finite element space. We can define an interpolation operator

$$\Pi_h : H^2(\Omega) \longrightarrow V_h$$

by the condition

$$\zeta = \Pi_h \zeta \quad \text{at the nodes of the Lagrange finite element}$$

Interpolation Error Estimates

$$\|\zeta - \Pi_h \zeta\|_{L_2(\Omega)} + h|\zeta - \Pi_h \zeta|_{H^1(\Omega)} \leq C h^m |\zeta|_{H^m(\Omega)} \qquad \forall\, \zeta \in H^m(\Omega)$$

$$(k+1 \geq m \geq 2)$$

The positive constant C depends only on the shapes of the elements and k (polynomial degree).

Sketch of Proof

Let $\hat{T}$ be the reference simplex and $\hat{\zeta} \in H^m(\hat{T})$ for $m \geq 2$.

$\Pi_{\hat{T}}\hat{\zeta} \in \mathbb{P}_k$ equals $\hat{\zeta}$ at all the nodes of the $\mathbb{P}_k$ Lagrange finite element.

Step 1 Show that $\|\Pi_{\hat{T}}\hat{\zeta}\|_{H^1(\hat{T})} \leq C\|\hat{\zeta}\|_{H^m(\hat{T})}$

Step 2 For any $p \in \mathbb{P}_k$

$$\begin{aligned}\|\hat{\zeta} - \Pi_{\hat{T}}\hat{\zeta}\|_{H^1(\hat{T})} &= \|(\hat{\zeta} - p) - \Pi_{\hat{T}}(\hat{\zeta} - p)\|_{H^1(\hat{T})} \\ &\leq C\|\hat{\zeta} - p\|_{H^m(\hat{T})}\end{aligned}$$

Step 3 Apply the Bramble-Hilbert lemma ($k + 1 \geq m$)

$$\inf_{p \in \mathbb{P}_k} \|\hat{\zeta} - p\|_{H^m(\hat{T})} \leq C|\hat{\zeta}|_{H^m(\hat{T})}$$

Step 4 Estimate for reference simplex

$$\|\hat{\zeta} - \Pi_{\hat{T}}\hat{\zeta}\|_{H^1(\hat{T})} \leq C|\hat{\zeta}|_{H^m(\hat{T})}$$

Step 5 Use a scaling argument to show that

$$\|\zeta - \Pi_T\zeta\|_{L_2(T)} + h_T|\zeta - \Pi_T\zeta|_{H^1(T)} \leq Ch_T^m|\zeta|_{H^m(T)}$$

- affine map $F : \hat{T} \longrightarrow T$ $\qquad (F\hat{x} = b + A\hat{x},\ A \in \mathbb{R}^{2\times 2},\ b \in \mathbb{R}^2)$
- $|\nabla F| \leq Ch_T$ and $|\nabla F^{-1}| \leq Ch_T^{-1}$
  ($C > 0$ depends only on the shape of T)
- $\zeta \in H^m(T)$ iff $\hat{\zeta} = \zeta \circ F \in H^m(\hat{T})$
- $\Pi_{\hat{T}}\hat{\zeta} = (\Pi_T\zeta) \circ F$

Concrete Error Estimates (diffusion, elasticity)

Let a second order SPD boundary value problem be solved by the Lagrange finite element method. We have

$$\begin{aligned} \|u - u_h\|_a &= \min_{v \in V_h} \|u - v\|_a \\ &\leq \|u - \Pi_h u\|_a \\ &\approx \|u - \Pi_h u\|_{H^1(\Omega)} \\ &\leq C h^{m-1} |u|_{H^m(\Omega)} \end{aligned}$$

Similar error estimates can be derived for finite element methods (Argyris, B-F-S) for fourth order SPD boundary value problems.

$$\|u - u_h\|_a \leq C h^{m-2} |u|_{H^m(\Omega)}$$

$$(k + 1 \geq m \geq 4)$$

Saddle Point Problems

Continuous Problem Find $(u,p) \in V \times Q$ such that

$$\begin{aligned} a(u,v) + b(v,p) &= F(v) \qquad \forall\, v \in V \\ b(u,q) &= G(q) \qquad \forall\, q \in Q \end{aligned}$$

Stability Conditions

- $a(\cdot,\cdot)$ (resp. $b(\cdot,\cdot)$) is a bounded bilinear form on $V \times V$ (resp. $V \times Q$).
- $a(v,v) \geq \alpha \|v\|_V^2$ if $b(v,q) = 0$ for all $q \in Q$
- $\inf\limits_{q \in Q} \sup\limits_{v \in V} \dfrac{b(v,q)}{\|v\|_V \|q\|_Q} \geq \beta > 0$ (Ladyzhenskaya-Babuška-Brezzi Condition)

Under these conditions the continuous saddle point problem has a unique solution.

A Priori Analysis

Finite Element Method

Find $(u_h, p_h) \in V_h \times Q_h \subset V \times Q$ such that

$$\begin{aligned} a(u_h, v) + b(v, p_h) &= F(v) \qquad \forall\, v \in V_h \\ b(u_h, q) \qquad\quad &= G(q) \qquad \forall\, q \in Q_h \end{aligned}$$

Stability Conditions

- $a(\cdot,\cdot)$ (resp. $b(\cdot,\cdot)$) is a bounded bilinear form on $V_h \times V_h$ (resp. $V_h \times Q_h$).
- $a(v,v) \geq \alpha_h \|v\|_V^2$ if $v \in V_h$ satisfies $b(v,q) = 0$ for all $q \in Q_h$
- $\inf\limits_{q \in Q_h} \sup\limits_{v \in V_h} \dfrac{b(v,q)}{\|v\|_V \|q\|_Q} \geq \beta_h > 0$

Under these conditions the discrete saddle point problem has a unique solution.

A Priori Analysis

The stability conditions

- $a(v,v) \geq \alpha_h \|v\|_V^2$ if $v \in V_h$ satisfies $b(v,q) = 0$ for all $q \in Q_h$
- $\inf\limits_{q \in Q_h} \sup\limits_{v \in V_h} \dfrac{b(v,q)}{\|v\|_V \|q\|_Q} \geq \beta_h > 0$

for the discrete saddle point problem do not follow from the corresponding conditions for the continuous saddle point problem

- $a(v,v) \geq \alpha \|v\|_V^2$ if $b(v,q) = 0$ for all $q \in Q$
- $\inf\limits_{q \in Q} \sup\limits_{v \in V} \dfrac{b(v,q)}{\|v\|_V \|q\|_Q} \geq \beta > 0$

Therefore the finite element pair V_h and Q_h must be chosen carefully.

Abstract Error Estimate

The discrete stability conditions imply the stability estimate

$$\|v\|_V + \|q\|_Q \leq \gamma_h \sup_{(w,r)\in V_h\times Q_h} \frac{a(v,w) + b(w,q) + b(v,r)}{\left(\|w\|_V^2 + \|r\|_Q^2\right)^{1/2}}$$

for all $(v,q) \in V_h \times Q_h$, where

$$\gamma_h = \frac{1}{\beta_h}\left(1 + \frac{C_a}{\beta_h}\right)\left(1 + \frac{C_a}{\alpha_h}\right)$$

and C_a is the bound of the bilinear form $a(\cdot,\cdot)$.

Galerkin Orthogonality

$$a(u - u_h, v) + b(v, p - p_h) + b(u - u_h, q) = 0 \qquad \forall\,(v,q) \in V_h \times Q_h$$

A Priori Analysis

For any $(v, q) \in V_h \times Q_h$

$$
\begin{aligned}
&\|u_h - v\|_V + \|p_h - q\|_Q \\
&\qquad \le \gamma_h \sup_{(w,r)\in V_h\times Q_h} \frac{a(u_h - v, w) + b(w, p_h - q) + b(u_h - v, r)}{\left(\|w\|_V^2 + \|r\|_Q^2\right)^{1/2}} \\
&\qquad = \gamma_h \sup_{(w,r)\in V_h\times Q_h} \frac{a(u - v, w) + b(w, p - q) + b(u - v, r)}{\left(\|w\|_V^2 + \|r\|_Q^2\right)^{1/2}} \\
&\qquad \le \gamma_h \left(C_a + C_b\right) \left(\|u - v\|_V + \|p - q\|_Q\right)
\end{aligned}
$$

C_b is the bound of $b(\cdot, \cdot)$

Therefore

$$
\begin{aligned}
&\|u - u_h\|_V + \|p - p_h\|_Q \\
&\qquad \le \left(\|u - v\|_V + \|p - q\|_Q\right) + \left(\|u_h - v\|_V + \|p_h - q\|_Q\right) \\
&\qquad \le \left[1 + \gamma_h \left(C_a + C_b\right)\right] \left(\|u - v\|_V + \|p - q\|_Q\right)
\end{aligned}
$$

A Priori Analysis

Hence

$$\|u-u_h\|_V+\|p-p_h\|_Q \leq \left[1+\gamma_h\left(C_a+C_b\right)\right] \inf_{(v,q)\in V_h\times Q_h} \left(\|u-v\|_V+\|p-q\|_Q\right)$$

Since

$$\gamma_h = \frac{1}{\beta_h}\left(1+\frac{C_a}{\beta_h}\right)\left(1+\frac{C_a}{\alpha_h}\right)$$

the mixed finite element method is quasi-optimal if

- α_h is bounded away from 0 (easy)
- β_h is bounded away from 0 (delicate)

Concrete error estimates can then be obtained by taking v and q to be appropriate interpolants of u and p.

Inverse Estimates

Since any two norms on a finite dimensional space are equivalent, a higher order (semi-) norm of a finite element function can be bounded by a lower order (semi-) norm.

$$
\begin{aligned}
|v|_{H^1(T)} &\leq C h_T^{-1} \|v\|_{L_2(T)} \\
\|v\|_{L_\infty(T)} &\leq C h_T^{-1} \|v\|_{L_2(T)} \\
|v|_{H^1(T)} &\leq C \|v\|_{L_\infty(T)}
\end{aligned}
$$

(C depends on the shape of the triangle and the degree of the polynomials involved.)

Such estimates are established by using
(i) the norm equivalence on the reference domain and
(ii) a scaling argument.

There are many other finite element estimates that can be proved by the same techniques:

$$\|v\|_{L_2(\partial T)} \leq C h_T^{-\frac{1}{2}} \|v\|_{L_2(T)}$$

We will refer to such estimates as discrete estimates.

We can obtain other (sub-optimal) error estimates using inverse estimates.

$$\begin{aligned}
\|u - u_h\|_{L_\infty(\Omega)} &\leq \|u - \Pi_h u\|_{L_\infty(\Omega)} + \|\Pi_h u - u_h\|_{L_\infty(\Omega)} \\
&\leq C\left(h^{k+1}\|u\|_{W_\infty^{k+1}(\Omega)} + \sum_{T\in\mathcal{T}_h} h_T^{-1}\|\Pi_h u - u_h\|_{L_2(T)}\right) \\
&\leq C\left(h^{k+1}\|u\|_{W_\infty^{k+1}(\Omega)} + \sum_{T\in\mathcal{T}_h} h_T^{-1}\left(\|\Pi_h u - u\|_{L_2(T)} + \|u - u_h\|_{L_2(T)}\right)\right) \\
&\leq C\left(h^{k+1}\|u\|_{W_\infty^{k+1}(\Omega)} + h^k |u|_{H^{k+1}(\Omega)}\right)
\end{aligned}$$

A Posteriori Analysis

The goal is to find a good estimate for the error of a finite element solution u_h using only u_h and the data of the boundary value problem.

We will focus on the diffusion problem solved by the $\mathbb{P}_k$ Lagrange finite element method.

Solution of Diffusion Problem $u \in H_0^1(\Omega)$ satisfies

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, dx \qquad \forall\, v \in H_0^1(\Omega)$$

Finite Element Solution $u_h \in V_h \, (\subset H_0^1(\Omega))$ satisfies

$$\int_\Omega \nabla u_h \cdot \nabla v \, dx = \int_\Omega f v \, dx \qquad \forall\, v \in V_h$$

$V_h = \mathbb{P}_k$ Lagrange finite element space associated with a triangulation $\mathcal{T}_h$ of Ω

A Posteriori Analysis

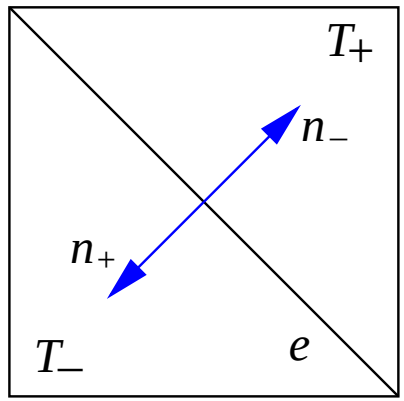
Notation Let v be a piecewise H^2 function.

$e =$ interior edge shared by $T_\pm$

$$[\![\partial v/\partial n]\!] = \nabla v_- \cdot \boldsymbol{n}_- + \nabla v_+ \cdot \boldsymbol{n}_+$$

$\boldsymbol{n}_\pm =$ unit outward normal of $T_\pm$

$v_\pm = v\big|_{T_\pm}$

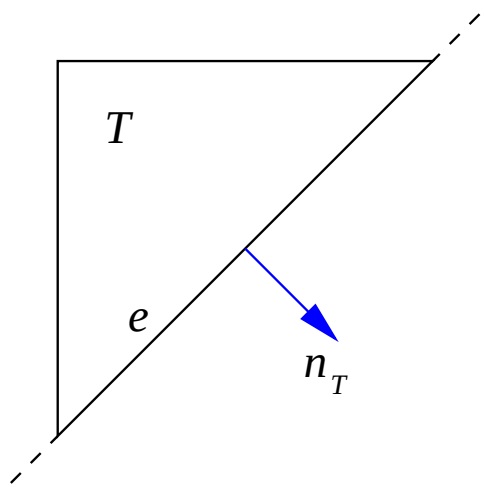


$e =$ boundary edge, $e \subset \partial T$

$$[\![\partial v/\partial n]\!] = \nabla v_T \cdot \boldsymbol{n}_T$$

$\boldsymbol{n}_T =$ unit outward normal of T

$v_T = v\big|_T$



A Residual Based Error Estimator

$$|u-u_h|_{H^1(\Omega)} = \sup_{v\in H_0^1(\Omega)} \frac{a(u-u_h,v)}{|v|_{H^1(\Omega)}} \qquad \text{(duality)}$$

$$a(w,v) = \int_\Omega \nabla w \cdot \nabla v \, dx$$

$$a(u-u_h,v) = a(u-u_h, v-\tilde{v}) \qquad \text{(for any } \tilde{v} \in V_h\text{)}$$

$$= \int_\Omega \nabla u \cdot \nabla (v-\tilde{v}) dx - \sum_{T\in\mathcal{T}_h} \int_T \nabla u_h \cdot \nabla (v-\tilde{v})\, dx$$

$$= \int_\Omega f(v-\tilde{v})dx + \sum_{T\in\mathcal{T}_h} \int_T (\Delta u_h)(v-\tilde{v})dx - \sum_{T\in\mathcal{T}_h} \int_{\partial T} \nabla u_h \cdot \boldsymbol{n}(v-\tilde{v})\, ds$$

$$= \sum_{T\in\mathcal{T}_h} \int_T (f+\Delta u_h)(v-\tilde{v})dx - \sum_{e\in\mathcal{E}_h^i} \int_e [\![\partial u_h/\partial n]\!](v-\tilde{v})ds$$

$\mathcal{E}_h^i =$ the set of the interior edges of $\mathcal{T}_h$

A Posteriori Analysis

Let $\tilde{v} \in V_h$ be a weak (or quasi-) interpolant of v such that

$$\sum_{T\in\mathcal{T}_h} h_T^{-2}\|v-\tilde{v}\|^2_{L_2(T)} + |\tilde{v}|^2_{H^1(\Omega)} \le C|v|^2_{H^1(\Omega)}$$

$$\begin{aligned}
a(u-u_h,v) &= \sum_{T\in\mathcal{T}_h}\int_T (f+\Delta u_h)(v-\tilde{v})dx - \sum_{e\in\mathcal{E}_h^i}\int_e [\![\partial u_h/\partial n]\!](v-\tilde{v})ds \\
&\le \Big(\sum_{T\in\mathcal{T}_h} h_T^2\|f+\Delta u_h\|^2_{L_2(T)}\Big)^{1/2}\Big(\sum_{T\in\mathcal{T}_h} h_T^{-2}\|v-\tilde{v}\|^2_{L_2(T)}\Big)^{1/2} \\
&\quad + \Big(\sum_{e\in\mathcal{E}_h^i} |e|\|[\![\partial u_h/\partial n]\!]\|^2_{L_2(e)}\Big)^{1/2}\Big(\sum_{e\in\mathcal{E}_h^i} |e|^{-1}\|v-\tilde{v}\|^2_{L_2(e)}\Big)^{1/2} \\
&\le C\Big(\sum_{T\in\mathcal{T}_h} h_T^2\|f+\Delta u_h\|^2_{L_2(T)} + \sum_{e\in\mathcal{E}_h^i} |e|\|[\![\partial u_h/\partial n]\!]\|^2_{L_2(e)}\Big)^{1/2}|v|_{H^1(\Omega)}
\end{aligned}$$

Therefore

$$|u-u_h|_{H^1(\Omega)} = \sup_{v\in H_0^1(\Omega)} \frac{a(u-u_h,v)}{|v|_{H^1(\Omega)}}$$

$$\leq C\Big(\sum_{T\in\mathcal{T}_h} h_T^2\|f+\Delta u_h\|_{L_2(T)}^2 + \sum_{e\in\mathcal{E}_h^i} |e|\|[\![\partial u_h/\partial n]\!]\|_{L_2(e)}^2\Big)^{1/2}$$

A Reliable Error Estimator

$$\eta_h = \Big(\sum_{T\in\mathcal{T}_h} h_T^2\|f+\Delta u_h\|_{L_2(T)}^2 + \sum_{e\in\mathcal{E}_h^i} |e|\|[\![\partial u_h/\partial n]\!]\|_{L_2(e)}^2\Big)^{1/2}$$

$$|u-u_h|_{H^1(\Omega)} \leq C\eta_h$$

$C>0$ depends only on the shapes of the triangles in $\mathcal{T}_h$ and k (polynomial degree).

Local Efficiency Estimates

Let $T \in \mathcal{T}_h$, $b \in \mathbb{P}_3(T)$ be the cubic bubble function that vanishes on ∂T and equals 1 at the center of T, and $\tilde{f}_T$ be the $L_2(T)$ projection of f in $\mathbb{P}_{k-1}(T)$.

$$\|\tilde{f}_T + \Delta u_h\|^2_{L_2(T)} \leq C \int_T (\tilde{f}_T + \Delta u_h)(\tilde{f}_T + \Delta u_h) b \, dx$$

$$= C\Big(\int_T (\tilde{f}_T - f)(\tilde{f}_T + \Delta u_h) b \, dx + \int_T (f + \Delta u_h)(\tilde{f}_T + \Delta u_h) b \, dx \Big)$$

$$\int_T (\tilde{f}_T - f)(\tilde{f}_T + \Delta u_h) b \, dx \leq C \|\tilde{f}_T - f\|_{L_2(T)} \|\tilde{f}_T + \Delta u_h\|_{L_2(T)}$$

$$\int_T (f + \Delta u_h)(\tilde{f}_T + \Delta u_h) b \, dx = \int_T \nabla(u - u_h) \cdot \nabla[(\tilde{f}_T + \Delta u_h) b] dx$$

$$\leq C |u - u_h|_{H^1(T)} h_T^{-1} \|\tilde{f}_T + \Delta u_h\|_{L_2(T)}$$

$$\|\tilde{f}_T + \Delta u_h\|^2_{L_2(T)} \leq C\big(\|\tilde{f}_T - f\|_{L_2(T)}\|\tilde{f}_T + \Delta u_h\|_{L_2(T)} + |u - u_h|_{H^1(T)} h_T^{-1} \|\tilde{f}_T + \Delta u_h\|_{L_2(T)}\big)$$

which implies

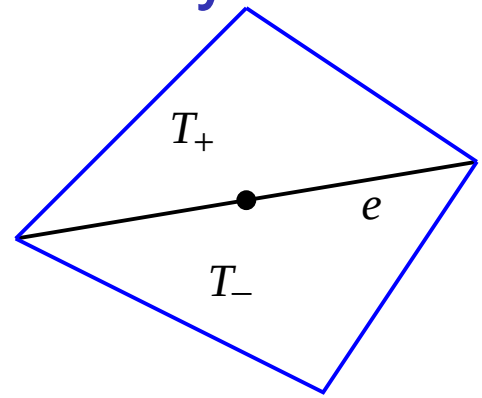
$$h_T\|\tilde{f}_T + \Delta u_h\|_{L_2(T)} \leq C\big(h_T\|\tilde{f}_T - f\|_{L_2(T)} + |u - u_h|_{H^1(T)}\big)$$

and hence

$$\begin{aligned} h_T\|f + \Delta u_h\|_{L_2(T)} &\leq h_T\big(\|f - \tilde{f}_T\|_{L_2(T)} + \|\tilde{f}_T + \Delta u_h\|_{L_2(T)}\big) \\ &\leq C\big(h_T\|f - \tilde{f}_T\|_{L_2(T)} + |u - u_h|_{H^1(T)}\big) \end{aligned}$$

A Posteriori Analysis

Let $e \in \mathcal{E}_h^i$ be the common edge of $T_\pm$.



$q \in \mathbb{P}_4$ is the bubble function that vanishes on the four outer edges of $T_- \cup T_+$ and equals 1 at the midpoint of e.

Let $p \in \mathbb{P}_{k-1}$ equal $[\![\partial u_h/\partial n]\!]$ on e and be constant on lines perpendicular to e.

$$\|[\![\partial u_h/\partial n]\!]\|^2_{L_2(e)} \le C \int_e [\![\partial u_h/\partial n]\!]^2 q \, ds$$

$$= C \int_e [\![\partial u_h/\partial n]\!](pq) ds$$

$$= C\Big(\int_{T_- \cup T_+} (\Delta u_h)(pq) dx + \int_{T_- \cup T_+} \nabla u_h \cdot \nabla(pq) \, dx \Big)$$

$$= C\Big(\int_{T_- \cup T_+} (f + \Delta u_h)(pq) dx + \int_{T_- \cup T_+} \nabla(u_h - u) \cdot \nabla(pq) \, dx \Big)$$

A Posteriori Analysis

$$\int_{T_-\cup T_+} (f+\Delta u_h)(pq)dx \le \|f+\Delta u_h\|_{L_2(T_-\cup T_+)}\|pq\|_{L_2(T_-\cup T_+)}$$
$$\le C\|f+\Delta u_h\|_{L_2(T_-\cup T_+)}|e|^{1/2}\|[\![\partial u_h/\partial n]\!]\|_{L_2(e)}$$

$$\int_{T_-\cup T_+} \nabla(u_h-u)\cdot\nabla(pq)\,dx \le |u-u_h|_{H^1(T_-\cup T_+)}|pq|_{H^1(T_-\cup T_+)}$$
$$\le C|u-u_h|_{H^1(T_-\cup T_+)}|e|^{-1}\|pq\|_{L_2(T_-\cup T_+)}$$
$$\le C|u-u_h|_{H^1(T_-\cup T_+)}|e|^{-(1/2)}\|[\![\partial u_h/\partial n]\!]\|_{L_2(e)}$$

$$\|[\![\partial u_h/\partial n]\!]\|^2_{L_2(e)} \le C\big(\|f+\Delta u_h\|_{L_2(T_-\cup T_+)}|e|^{1/2}\|[\![\partial u_h/\partial n]\!]\|_{L_2(e)}$$
$$+|u-u_h|_{H^1(T_-\cup T_+)}|e|^{-(1/2)}\|[\![\partial u_h/\partial n]\!]\|_{L_2(e)}\big)$$

A Posteriori Analysis

After cancellation,

$$
\begin{aligned}
|e|^{1/2}\|[\![\partial u_h/\partial n]\!]\|_{L_2(e)} &\le C\big(|e|\|f+\Delta u_h\|_{L_2(T_-\cup T_+)} + |u-u_h|_{H^1(T_-\cup T_+)}\big)\\
&\le C\big(h_{T_-}\|f+\Delta u_h\|_{L_2(T_-)} + h_{T_+}\|f+\Delta u_h\|_{L_2(T_+)}\\
&\qquad + |u-u_h|_{H^1(T_-\cup T_+)}\big)
\end{aligned}
$$

Recall

$$
h_T\|f+\Delta u_h\|_{L_2(T)} \le C\big(h_T\|f-\tilde{f}_T\|_{L_2(T)} + |u-u_h|_{H^1(T)}\big)
$$

Hence

$$
|e|^{1/2}\|[\![\partial u_h/\partial n]\!]\|_{L_2(e)} \le C\Big(\sum_{T\in\mathcal{T}_e} h_T\|f-\tilde{f}_T\|_{L_2(T)} + \sum_{T\in\mathcal{T}_e}|u-u_h|_{H^1(T)}\Big)
$$

$\mathcal{T}_e=\{T_-,T_+\}$ is the set of the triangles in $\mathcal{T}_h$ that share the common edge e.

Efficiency of the Error Estimator

$$h_T \|f + \Delta u_h\|_{L_2(T)} \leq C\big(h_T \|f - \tilde{f}_T\|_{L_2(T)} + |u - u_h|_{H^1(T)}\big)$$

$$|e|^{1/2} \|[\![\partial u_h/\partial n]\!]\|_{L_2(e)} \leq C\Big(\sum_{T\in\mathcal{T}_e} h_T \|f - \tilde{f}_T\|_{L_2(T)} + \sum_{T\in\mathcal{T}_e} |u - u_h|_{H^1(T)}\Big)$$

$$\begin{aligned}\eta_h^2 &= \sum_{T\in\mathcal{T}_h} h_T^2 \|f + \Delta u_h\|_{L_2(T)}^2 + \sum_{e\in\mathcal{E}_h^i} |e| \|[\![\partial u_h/\partial n]\!]\|_{L_2(e)}^2 \\ &\leq C\Big(|u - u_h|_{H^1(\Omega)}^2 + \sum_{T\in\mathcal{T}_h} h_T^2 \|f - \tilde{f}_T\|_{L_2(T)}^2\Big)\end{aligned}$$

or

$$\eta_h \leq C\big[|u - u_h|_{H^1(\Omega)} + \mathrm{Osc}(f)\big]$$

$$\mathrm{Osc}(f) = \Big(\sum_{T\in\mathcal{T}_h} h_T^2 \|f - \tilde{f}_T\|_{L_2(T)}^2\Big)^{1/2}$$

Reliability and efficiency together

$$C_1 |u - u_h|_{H^1(\Omega)} \leq \eta_h \leq C_2 \left[|u - u_h|_{H^1(\Omega)} + \mathrm{Osc}(f) \right]$$

means that up to a higher order term η_h is both an upper bound and a lower bound for the true error $|u - u_h|_{H^1(\Omega)}$.

$$\mathrm{Osc}(f) = \Big(\sum_{T \in \mathcal{T}_h} h_T^2 \|f - \tilde{f}_T\|_{L_2(T)}^2 \Big)^{1/2} = O(h^{k+1})$$

if $f \in H^k(\Omega)$ ($\tilde{f}_T \in \mathbb{P}_{k-1}(T)$ is the $L_2(T)$ projection of f.)

The error estimator η_h can be used in the design of an adaptive algorithm:

solve $\longrightarrow$ estimate $\longrightarrow$ mark $\longrightarrow$ refine

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   A Posteriori Error Estimation in Finite Element Analysis, Wiley-Interscience, 2000.

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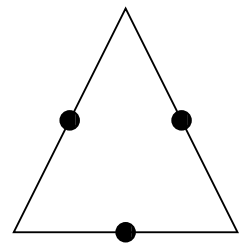
Variational Crimes

A *variational crime* is committed if the finite element space V_h is not a subspace of the space V where the continuous problem is posed.

Example Crouzeix-Raviart nonconforming $\mathbb{P}_1$ finite element

$\mathcal{K} =$ triangle

$\mathcal{P} = \mathbb{P}_1$



$\mathcal{T}_h =$ triangulation of Ω, $V_h =$ finite element space

$v \in V_h$ if and only if

- $v\big|_T \in \mathbb{P}_1(T)$ for all $T \in \mathcal{T}_h$
- v is continuous at the interior midpoints
- v vanishes at the boundary midpoints

Nonconforming $\mathbb{P}_1$ FEM for the Diffusion Problem

Find $u_h \in V_h$ such that

$$a_h(u_h, v) = \int_\Omega fv \, dx \qquad \forall\, v \in V_h$$

where

$$a_h(w, v) = \sum_{T \in \mathcal{T}_h} \int_T \nabla w \cdot \nabla v \, dx$$

Since $V_h \not\subset V = H_0^1(\Omega)$, we no longer have Galerkin orthogonality. This inconsistency is quantified by

$$\sup_{v \in V_h \setminus \{0\}} \frac{a_h(u - u_h, v)}{\|v\|_{a_h}}$$

where u is the solution of the diffusion problem and $\|v\|_{a_h} = \sqrt{a_h(v, v)}$.

Variational Crimes

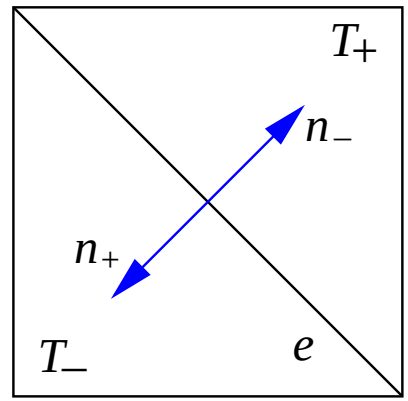
Notation Let v be a piecewise H^1 function.

$e =$ interior edge shared by $T_\pm$

$$[\![v]\!] = v_- \boldsymbol{n}_- + v_+ \boldsymbol{n}_+$$

$\boldsymbol{n}_\pm =$ unit outward normal of $T_\pm$

$v_\pm = v\big|_{T_\pm}$

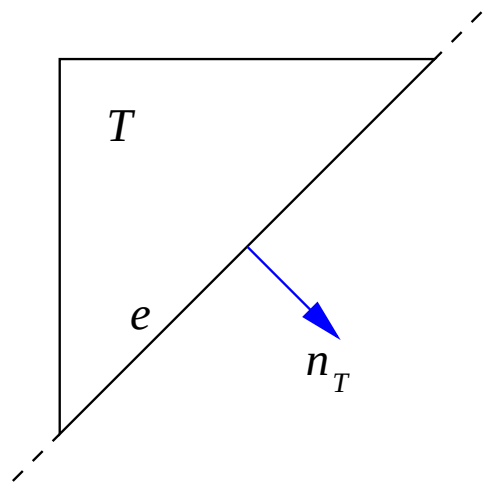


$e =$ boundary edge, $e \subset \partial T$

$$[\![v]\!] = v_T \boldsymbol{n}_T$$

$\boldsymbol{n}_T =$ unit outward normal of T

$v_T = v\big|_T$



Variational Crimes

For any $v \in V_h$,

$$
\begin{aligned}
a_h(u, v) &= \sum_{T \in \mathcal{T}_h} \int_T \nabla u \cdot \nabla v \, dx \\
&= -\sum_{T \in \mathcal{T}_h} \int_T (\Delta u) v \, ds + \sum_{T \in \mathcal{T}_h} \int_{\partial T} \nabla u \cdot \boldsymbol{n} \, v \, ds \\
&= \int_\Omega f v \, dx + \sum_{e \in \mathcal{E}_h} \int_e \nabla u \cdot [\![v]\!] ds
\end{aligned}
$$

$$
a_h(u_h, v) = \int_\Omega f v \, dx
$$

$$
a_h(u - u_h, v) = \sum_{e \in \mathcal{E}_h} \int_e \nabla u \cdot [\![v]\!] ds
$$

Variational Crimes

Let $\Pi_e^0 \nabla u = \dfrac{1}{|e|} \displaystyle\int_e \nabla u \, ds.$

$$
\begin{aligned}
a_h(u - u_h, v) &= \sum_{e \in \mathcal{E}_h} \int_e \nabla u \cdot [\![v]\!] ds \\
&= \sum_{e \in \mathcal{E}_h} \int_e (\nabla u - \Pi_e^0 \nabla u) \cdot [\![v]\!] ds \\
&\leq \Big(\sum_{e \in \mathcal{E}_h} |e| \| \nabla u - \Pi_e^0 \nabla u \|_{L_2(e)}^2 \Big)^{1/2} \Big(\sum_{e \in \mathcal{E}_h} |e|^{-1} \| [\![v]\!] \|_{L_2(e)}^2 \Big)^{1/2} \\
&\leq Ch |u|_{H^2(\Omega)} \| v \|_{a_h}
\end{aligned}
$$

$$
\sup_{v \in V_h \setminus \{0\}} \frac{a_h(u - u_h, v)}{\| v \|_{a_h}} \leq Ch |u|_{H^2(\Omega)}
$$

An Abstract Error Estimate

For any $v \in V_h$,

$$\begin{aligned}
\|u - u_h\|_{a_h} &\leq \|u - v\|_{a_h} + \|v - u_h\|_{a_h} \\
&\leq \|u - v\|_{a_h} + \sup_{w \in V_h \setminus \{0\}} \frac{a_h(v - u_h, w)}{\|w\|_{a_h}} \\
&\leq \|u - v\|_{a_h} + \sup_{w \in V_h \setminus \{0\}} \frac{a_h(v - u, w) + a_h(u - u_h, w)}{\|w\|_{a_h}} \\
&\leq 2\|u - v\|_{a_h} + \sup_{w \in V_h \setminus \{0\}} \frac{a_h(u - u_h, w)}{\|w\|_{a_h}}
\end{aligned}$$

$$\|u - u_h\|_{a_h} \leq 2 \inf_{v \in V_h} \|u - v\|_{a_h} + \sup_{w \in V_h \setminus \{0\}} \frac{a_h(u - u_h, w)}{\|w\|_{a_h}}$$

Variational Crimes

Concrete Error Estimate

$$\inf_{v \in V_h} \|u - v\|_{a_h} \leq \|u - \Pi_h u\|_{a_h} \leq Ch|u|_{H^2(\Omega)}$$

$$\sup_{w \in V_h \setminus \{0\}} \frac{a_h(u - u_h, w)}{\|w\|_{a_h}} \leq Ch|u|_{H^2(\Omega)}$$

$$\begin{aligned} \|u - u_h\|_{a_h} &\leq 2 \inf_{v \in V_h} \|u - v\|_{a_h} + \sup_{w \in V_h \setminus \{0\}} \frac{a_h(u - u_h, w)}{\|w\|_{a_h}} \\ &\leq Ch|u|_{H^2(\Omega)} \end{aligned}$$

The nonconforming $\mathbb{P}_1$ finite element method can be applied to the Stokes problem.

M. Crouzeix and P.-A. Raviart

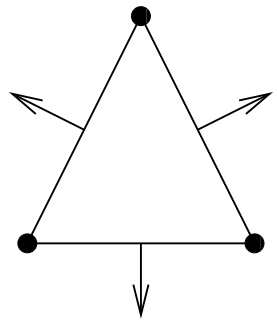
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Variational Crimes

Example Morley element

$$\mathcal{K} = \text{triangle}$$

$$\mathcal{P} = \mathbb{P}_2$$



$\mathcal{T}_h =$ triangulation of Ω, $\quad V_h =$ finite element space

$v \in V_h$ if and only if

- $v\big|_T \in \mathbb{P}_2(T)$ for all $T \in \mathcal{T}_h$
- v is continuous at the vertices
- $\partial v/\partial n$ is continuous at the midpoints
- v vanishes at the vertices on $\partial\Omega$
- $\partial v/\partial n$ vanishes at the midpoints on $\partial\Omega$

Variational Crimes

Morley FEM for the Biharmonic Problem

Find $u_h \in V_h$ such that

$$a_h(u_h, v) = \int_\Omega fv \, dx \qquad \forall\, v \in V_h$$

where

$$a_h(w, v) = \sum_{T \in \mathcal{T}_h} \int_T \nabla^2 w : \nabla^2 v \, dx$$

Error Estimate

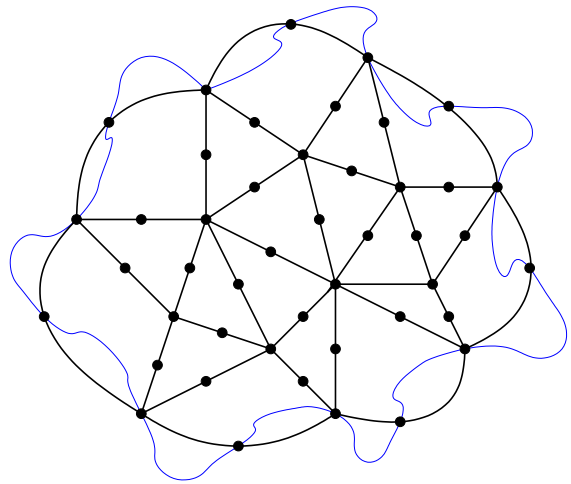
$$\begin{aligned} \|u - u_h\|_{a_h} &\leq 2 \inf_{v \in V_h} \|u - v\|_{a_h} + \sup_{w \in V_h \setminus \{0\}} \frac{a_h(u - u_h, w)}{\|w\|_{a_h}} \\ &\leq Ch|u|_{H^3(\Omega)} \end{aligned}$$

Variational Crimes

Example isoparametric finite element

Ω is a smooth domain.

Ω_h is an approximation of Ω consisting of curvilinear triangles.



Properties of $\mathcal{T}_h$

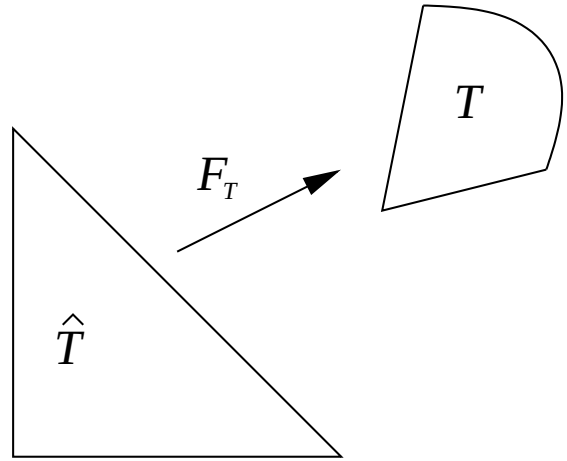
- The vertices of the curved edges belong to $\partial\Omega$.
- Each triangle $T \in \mathcal{T}_h$ is the image of the reference simplex $\hat{T}$ under an invertible polynomial map F_T of degree $\leq k$, where k is the degree of the polynomial shape functions.
- All the nodes of the finite elements belong to $\bar{\Omega}$ and the nodes on the boundary of Ω_h also belong to $\partial\Omega$.

Variational Crimes

Isoparametric $\mathbb{P}_k$ Lagrange Finite Element Space

$v \in V_h$ if and only if $\qquad (v_T = v\big|_T)$

- $v_T \circ F_T \in \mathbb{P}_k(\hat{T})$ for all $T \in \mathcal{T}_h$
- $v \in C(\bar{\Omega}_h)$
- v vanishes on $\partial\Omega_h$



Interpolation error estimates can be extended to isoparametric finite elements.

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Variational Crimes

Isoparametric FEM for the Diffusion Problem

Find $u_h \in V_h$ such that

$$a_h(u_h, v) = \int_{\Omega_h} (\Pi_h f) v \, dx \qquad \forall\, v \in V_h$$

where

$$a_h(w, v) = \int_{\Omega_h} \nabla w \cdot \nabla v \, dx$$

and $\Pi_h f$ is the nodal interpolant of f.

We assume that f (and hence u) is sufficiently smooth so that $\Pi_h f$ is well-defined and the full approximation power of V_h can be realized. For the error analysis we also assume that u has been extended to $\mathbb{R}^2$ (still denoted by u).

Variational Crimes

Abstract Error Estimate

$$\|u - u_h\|_{a_h} \leq 2 \inf_{v \in V_h} \|u - v\|_{a_h} + \sup_{w \in V_h \setminus \{0\}} \frac{a_h(u - u_h, w)}{\|w\|_{a_h}}$$

Approximation Property

$$\inf_{v \in V_h} \|u - v\|_{a_h} \leq \|u - \Pi_h u\|_{a_h} \leq C h^k \|u\|_{H^{k+1}(\Omega)}$$

Inconsistency Error

$$\begin{aligned}
a_h(u - u_h, w) &= a_h(u, w) - a_h(u_h, w) \\
&= \int_{\Omega_h} \nabla u \cdot \nabla w \, dx - \int_{\Omega_h} (\Pi_h f) w \, dx \\
&= -\int_{\Omega_h} (\Delta u) w \, dx + \int_{\Omega_h} (\Pi_h \Delta u) w \, dx \\
&\leq \|\Delta u - \Pi_h \Delta u\|_{L_2(\Omega_h)} \|w\|_{L_2(\Omega_h)} \leq C h^{k+1} \|u\|_{H^{k+3}(\Omega)} \|w\|_{a_h}
\end{aligned}$$

Variational Crimes

Concrete Error Estimate

$$|u - u_h|_{H^1(\Omega_h)} = \|u - u_h\|_{a_h} \leq Ch^k \|u\|_{H^{k+3}(\Omega)}$$

It is possible (cf. Lenoir 1986) to construct a piecewise smooth map $\Phi_h : \Omega \longrightarrow \Omega_h$ such that the Jacobian J_{Φ_h} satisfies

$$|J_{\Phi_h} - I| \leq Ch^k$$

Then $\hat{u}_h = u_h \circ \Phi_h \in H_0^1(\Omega)$ satisfies

$$|u - \hat{u}_h|_{H^1(\Omega)} \leq Ch^k$$

L_2 Error Estimate

Let $u_h \in V_h$ be a Lagrange finite element approximation of the solution u of the diffusion problem. The L_2 error $\|u - u_h\|_{L_2(\Omega)}$ can be estimated using the Aubin-Nitsche duality argument.

$$\|u - u_h\|_{L_2(\Omega)} = \sup_{\phi \in L_2(\Omega) \setminus \{0\}} \frac{(\phi, u - u_h)_{L_2(\Omega)}}{\|\phi\|_{L_2(\Omega)}}$$

Given any $\phi \in L_2(\Omega)$, let $\zeta \in H_0^1(\Omega)$ be the solution of

$$a(v, \zeta) = \int_\Omega \phi v \, dx \qquad \forall\, v \in H_0^1(\Omega)$$

where

$$a(w, v) = \int_\Omega \nabla w \cdot \nabla v \, dx$$

i.e., ζ is the solution of the diffusion problem with ϕ as the right-hand side.

Miscellaneous Topics

$$\begin{aligned}(\phi, u-u_h)_{L_2(\Omega)} &= a(u-u_h, \zeta) \\ &= a(u-u_h, \zeta - \Pi_h \zeta) \\ &\leq \|u-u_h\|_a \|\zeta - \Pi_h \zeta\|_a\end{aligned}$$

If Ω is convex, then

$$\|\zeta\|_{H^2(\Omega)} \leq C_\Omega \|\phi\|_{L_2(\Omega)}$$

and hence

$$(\phi, u-u_h) \leq \|u-u_h\|_a \big(Ch|\zeta|_{H^2(\Omega)}\big) \leq \|u-u_h\|_a \big(Ch\|\phi\|_{L_2(\Omega)}\big)$$

$$\|u-u_h\|_{L_2(\Omega)} = \sup_{\phi \in L_2(\Omega)\setminus\{0\}} \frac{(\phi, u-u_h)_{L_2(\Omega)}}{\|\phi\|_{L_2(\Omega)}} \leq Ch\|u-u_h\|_a$$

FEM with Numerical Integration

Find $u_h \in V_h$ such that

$$\int_\Omega \nabla u_h \cdot \nabla v \, dx = (\text{num}) \int_\Omega f v \, dx \qquad \forall\, v \in V_h$$

Abstract Error Estimate

$$|u - u_h|_{H^1(\Omega)} \leq C \Bigg(\inf_{v \in V_h} |u - v|_{H^1(\Omega)} + \sup_{w \in H_0^1(\Omega) \setminus \{0\}} \frac{\int_\Omega f w \, dx - (\text{num}) \int_\Omega f w \, dx}{|v|_{H^1(\Omega)}} \Bigg)$$

Miscellaneous Topics

Nonhomogeneous Boundary Conditions

$$\begin{aligned} -\Delta u &= f \qquad \text{in } \Omega \\ u &= g \qquad \text{on } \partial\Omega \end{aligned}$$

Weak Formulation

Find $u \in H^1(\Omega)$ such that

$$\begin{aligned} \int_\Omega \nabla u \cdot \nabla v \, dx &= \int_\Omega f v \, dx \qquad \forall\, v \in H_0^1(\Omega) \\ u &= g \qquad \text{on } \partial\Omega \end{aligned}$$

$V_h (\subset H^1(\Omega)) = \mathbb{P}_k$ Lagrange finite element space

$\mathring{V}_h (\subset H_0^1(\Omega)) =$ subspace of V_h whose members vanish on $\partial\Omega$

Miscellaneous Topics

Finite Element Method

Find $u_h \in V_h$ such that

$$\int_\Omega \nabla u_h \cdot \nabla v \, dx = \int_\Omega f v \, dx \qquad \forall\, v \in \mathring{V}_h$$
$$u_h = \Pi_h g \qquad \text{on } \partial\Omega$$

Galerkin Orthogonality

$$\int_\Omega \nabla(u - u_h) \cdot \nabla v \, dx = 0 \qquad \forall\, v \in \mathring{V}_h$$

Error Estimate

$$|u-u_h|_{H^1(\Omega)} = \inf_{\substack{v \in V_h \\ v|_{\partial\Omega} = \Pi_h g}} |u-v|_{H^1(\Omega)} \leq |u-\Pi_h u|_{H^1(\Omega)} \leq Ch^k |u|_{H^{k+1}(\Omega)}$$

Nonsymmetric Coercive Problems

$$\begin{aligned} -\Delta u + (\boldsymbol{b} \cdot \nabla) u + cu &= f \qquad \text{in } \Omega \\ u &= 0 \qquad \text{on } \partial\Omega \end{aligned}$$

$c \geq 0$ and $\boldsymbol{b} \in \mathbb{R}^2$

Weak Formulation

Find $u \in H_0^1(\Omega)$ such that

$$a(u, v) = \int_\Omega fv \, dx \qquad \forall\, v \in H_0^1(\Omega)$$

where

$$a(w, v) = \int_\Omega \big(\nabla w \cdot \nabla v + (\boldsymbol{b} \cdot \nabla u) v + cuv \big) dx$$

Miscellaneous Topics

Properties of $a(\cdot,\cdot)$

$$a(w,v) \le \alpha |w|_{H^1(\Omega)} |v|_{H^1(\Omega)} \qquad \forall\, v, w \in H_0^1(\Omega)$$
$$a(v,v) \ge \beta |v|^2_{H^1(\Omega)} \qquad \forall\, v \in H_0^1(\Omega)$$

The boundary value problem is well-posed by the Lax-Milgram Theorem.

Finite Element Method

Find $u_h \in V_h$ such that

$$a(u_h, v) = \int_\Omega f v \, dx \qquad \forall\, v \in V_h$$

where $V_h (\subset H_0^1(\Omega))$ is the $\mathbb{P}_k$ Lagrange finite element space.

Galerkin Orthogonality

$$a(u - u_h, v) = 0 \qquad \forall\, v \in V_h$$

Error Estimate

$$\begin{aligned} |u - u_h|^2_{H^1(\Omega)} &\leq \frac{1}{\beta} a(u - u_h, u - u_h) \\ &\leq \frac{1}{\beta} a(u - u_h, u - \Pi_h u) \\ &\leq \frac{\alpha}{\beta} |u - u_h|_{H^1(\Omega)} |u - \Pi_h u|_{H^1(\Omega)} \end{aligned}$$

$$|u - u_h|_{H^1(\Omega)} \leq \frac{\alpha}{\beta} |u - \Pi_h u|_{H^1(\Omega)} \leq C h^k |u|_{H^{k+1}(\Omega)}$$

Indefinite Problems

$$\begin{aligned} -\Delta u + (\boldsymbol{b} \cdot \nabla) u + cu &= f \qquad \text{in } \Omega \\ u &= 0 \qquad \text{on } \partial\Omega \end{aligned}$$

$c < 0$ and $\boldsymbol{b} \in \mathbb{R}^2$

We assume that this problem is uniquely solvable.

Weak Formulation Find $u \in H_0^1(\Omega)$ such that

$$a(u, v) = \int_\Omega fv \, dx \qquad \forall\, v \in H_0^1(\Omega)$$

where

$$a(w, v) = \int_\Omega \big(\nabla w \cdot \nabla v + (\boldsymbol{b} \cdot \nabla u) v + cuv \big) dx$$

Finite Element Method

Find $u_h \in V_h$ such that

$$a(u_h, v) = \int_\Omega fv \, dx \qquad \forall\, v \in V_h$$

where $V_h (\subset H_0^1(\Omega))$ is a Lagrange finite element space.

In this case the discrete problem is *not* necessarily well-posed.

The Approach of Schatz

Assume that u_h exists and derive *a priori* error estimates for $|u - u_h|_{H^1(\Omega)}$ using Gårding's inequality and a duality argument. This then implies the discrete problem is well-posed for h sufficiently small.

Miscellaneous Topics

Gårding's Inequality

$$\frac{1}{2}|v|^2_{H^1(\Omega)} \leq a(v,v) + K\|v\|^2_{L_2(\Omega)} \qquad \forall\, v \in H^1(\Omega)$$

for a sufficiently large positive constant K

Duality Argument

For any $\phi \in L_2(\Omega)$, there exists a unique $\zeta \in H^1_0(\Omega)$ such that

$$a(v,\zeta) = \int_\Omega \phi v \, dx \qquad \forall\, v \in H^1_0(\Omega)$$

i.e., ζ is the solution of the adjoint boundary value problem

$$\begin{aligned} -\Delta\zeta - \nabla\cdot(\zeta \boldsymbol{b}) + c\zeta &= \phi \qquad \text{in } \Omega \\ \zeta &= 0 \qquad \text{on } \partial\Omega \end{aligned}$$

For simplicity we assume Ω is convex, in which case

$$\|\zeta\|_{H^2(\Omega)} \leq C_\Omega \|\phi\|_{L_2(\Omega)}$$

and we obtain by duality (assuming that u_h exists)

$$\|u - u_h\|_{L_2(\Omega)} \leq Ch|u - u_h|_{H^1(\Omega)}$$

$$\begin{aligned}
\frac{1}{2}|u - u_h|^2_{H^1(\Omega)} &\leq a(u - u_h, u - u_h) + K\|u - u_h\|^2_{L_2(\Omega)} \\
&\leq a(u - u_h, u - \Pi_h u) + Ch^2|u - u_h|^2_{H^1(\Omega)} \\
&\leq C\big(|u - u_h|_{H^1(\Omega)}|u - \Pi_h u|_{H^1(\Omega)} + h^2|u - u_h|^2_{H^1(\Omega)}\big)
\end{aligned}$$

and hence, for h sufficiently small,

$$|u - u_h|_{H^1(\Omega)} \leq C|u - \Pi_h u|_{H^1(\Omega)} \leq Ch^k|u|_{H^{k+1}(\Omega)}$$

In the case where $f = 0$ (and hence $u = 0$), we have

$$|u_h|_{H^1(\Omega)} \leq 0 \Longrightarrow u_h = 0$$

i.e., the only solution of the homogeneous problem

$$a(u_h, v) = 0 \qquad \forall\, v \in V_h$$

is the trivial solution. Therefore the discrete problem is nonsingular.

Hence the conditional error estimates become true estimates provided h is sufficiently small:

$$|u - u_h|_{H^1(\Omega)} \leq Ch^k |u|_{H^{k+1}(\Omega)}$$
$$\|u - u_h\|_{L_2(\Omega)} \leq Ch|u - u_h|_{H^1(\Omega)}$$

Properties of Finite Element Equations

We can represent the bilinear form $a_h(\cdot,\cdot)$ on $V_h \times V_h$ by the linear map $A_h : V_h \longrightarrow V_h'$ such that

$$\langle A_h w, v\rangle = a_h(w,v) \qquad \forall\, v, w \in V_h$$

where $\langle\cdot,\cdot\rangle$ is the canonical bilinear form on $V_h' \times V_h$.

Then the finite element equation

$$a_h(u_h, v) = F(v) \qquad \forall\, v \in V_h$$

can be written as

$$A_h u_h = \phi_h$$

where $\phi_h \in V_h'$ is defined by

$$\langle \phi_h, v\rangle = F(v) \qquad \forall\, v \in V_h$$

Miscellaneous Topics

The natural nodal basis of V_h consists of finite element functions whose dofs are all zeros except one that takes the value of 1. With respect to the natural nodal basis and the dual basis of V_h', we can represent the finite element equation as a linear system with coefficient matrix $\mathbf{A}_h$, unknown vector $\mathbf{x}_h$ and right-hand side vector $\mathbf{f}_h$:

$$\mathbf{A}_h \mathbf{x}_h = \mathbf{f}_h$$

$\mathbf{A}_h$ is known as the stiffness matrix.

Since the natural nodal basis functions are locally supported, the stiffness matrix is sparse, i.e., most of the components along any row or column of $\mathbf{A}$ equal 0.

Miscellaneous Topics

For a SPD problem, the stiffness matrix $\mathbf{A}_h$ is SPD and we can define the condition number $\kappa(\mathbf{A}_h)$ by

$$\kappa(\mathbf{A}_h) = \frac{\lambda_{\max}(\mathbf{A}_h)}{\lambda_{\min}(\mathbf{A}_h)}$$

where $\lambda_{\max}(\mathbf{A}_h)$ and $\lambda_{\min}(\mathbf{A}_h)$ are the maximum and minimum eigenvalues of $\mathbf{A}_h$.

For a second order problem discretized by a Lagrange finite element method on a quasi-uniform triangulation $\mathcal{T}_h$, it follows from the Rayleigh quotient formula and standard inverse estimates that

$$\lambda_{\max}(\mathbf{A}_h) = \max_{\mathbf{x}\in\mathbb{R}^2\setminus\{\mathbf{0}\}} \frac{\mathbf{x}^t\mathbf{A}_h\mathbf{x}}{\mathbf{x}^t\mathbf{x}} \approx \max_{v\in V_h\setminus\{0\}} \frac{a_h(v,v)}{h^{-2}\|v\|^2_{L_2(\Omega)}} \leq C_{\max}$$

In the other direction, it follows from the Rayleigh quotient formula and a Poincaré-Friedrichs inequality that

$$\lambda_{\min}(\mathbf{A}_h) = \min_{\mathbf{x}\in\mathbb{R}^2\setminus\{\mathbf{0}\}} \frac{\mathbf{x}^t\mathbf{A}_h\mathbf{x}}{\mathbf{x}^t\mathbf{x}} \approx \min_{v\in V_h\setminus\{0\}} \frac{a_h(v,v)}{h^{-2}\|v\|^2_{L_2(\Omega)}} \geq C_{\min}h^2$$

Condition Number Estimate

$$\kappa(\mathbf{A}_h) = \frac{\lambda_{\max}(\mathbf{A}_h)}{\lambda_{\min}(\mathbf{A}_h)} \leq Ch^{-2}$$

The condition number estimate is sharp.

Similar estimates can be developed for higher order problems.